

Operations Research: Algebra and Proofs

Review sheet and proof problem set for Lectures 1 to 5

Notation, ten algebraic moves, eleven proof patterns, the theorem map, and 18 proof problems with 18 true/false statements

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1 The Setting and the Notation

Everything in Lectures 2 to 5 happens in one of two settings. Keep both in view; most proofs move between them.

Equational form	$\max c^\top x$ s.t. $Ax = b, x \geq 0$, with $A \in \mathbb{R}^{m \times n}$ of full row rank m ; feasible set
(E)	$P = \{x : Ax = b, x \geq 0\}$.
Its dual (E*)	$\min b^\top p$ s.t. $A^\top p \geq c, p \in \mathbb{R}^m$ free (equality rows give free dual variables).
Canonical pair	(P) $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$ and (D) $\min b^\top y$ s.t. $A^\top y \geq c, y \geq 0$.
Columns and rows	A_j is column j of A ; $a_i^\top$ is row i . For an index set S : x_S, c_S are the entries in S and A_S the columns in S . $\text{supp}(x) = \{j : x_j \neq 0\}$.
Basis	m indices $B(1), \dots, B(m)$ with $B = [A_{B(1)} \cdots A_{B(m)}]$ invertible; N = the other indices. The same letter B names the index list and the matrix.
Basic solution	$x_N = 0, x_B = B^{-1}b$. Feasible (a BFS) if $x_B \geq 0$; nondegenerate if $x_B > 0$.
Multipliers, reduced costs	$p^\top = c_B^\top B^{-1}$ (so $B^\top p = c_B$) and $r = c - A^\top p$, i.e. $r_j = c_j - p^\top A_j$.
Basic direction	For $j \in N$: $d_j = 1, d_i = 0$ for the other $i \in N$, and $d_B = -B^{-1}A_j$. Write $u = B^{-1}A_j = -d_B$.
Tableau of B	$x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj}x_j$ and $z = z_0 + \sum_{j \in N} r_jx_j$, with $\beta = B^{-1}b$, $q_{\cdot j} = -B^{-1}A_j$, $z_0 = c_B^\top B^{-1}b$.
Optimal basis	$B^{-1}b \geq 0$ (primal feasible) and $r \leq 0$ (optimality, for a maximisation).

Three sign traps

- The tableau shows $-B^{-1}A_j$, not $B^{-1}A_j$, because the basic variables are isolated on the left.
- For a *maximisation*, optimality is $r \leq 0$. For a minimisation it is $r \geq 0$.
- p is free in (E*) but $y \geq 0$ in (D). The sign comes from the slack columns: in the canonical case the slack s_i has reduced cost $-p_i$, so $r \leq 0$ includes $p \geq 0$.

2 Ten Algebraic Moves

Almost every line of every proof in the course is one of these moves. Learn them as reflexes, with the one-line reason.

M1. Split a product by columns

$Ax = \sum_{j=1}^n x_j A_j$. So for any partition of the indices, $Ax = A_S x_S + A_{\bar{S}} x_{\bar{S}}$, and in particular

$$Ax = Bx_B + A_N x_N, \quad c^\top x = c_B^\top x_B + c_N^\top x_N.$$

Use: every argument that “fixes x_N and solves for x_B ”. Row view: $(Ax)_i = a_i^\top x$.

M2. A scalar equals its transpose

$y^\top Ax = (A^\top y)^\top x = x^\top A^\top y$. *Use:* moving A from the x side to the y side; this is the whole of “rows become columns” in duality. In particular, if $Ax = b$ then $b^\top p = (A^\top p)^\top x$ for every p .

M3. Signs of inner products

If $u \leq v$ componentwise and $w \geq 0$, then $w^\top u \leq w^\top v$ (each term satisfies $w_i u_i \leq w_i v_i$). If $u, w \geq 0$ then $w^\top u \geq 0$. A sum of nonnegative numbers that equals 0 has every term equal to 0. *Use:* weak duality, complementary slackness, uniqueness, every optimality test.

M4. What B^{-1} does to the basic columns

$B^{-1}A_{B(k)} = e_k$, because $B^{-1}B = I$. Hence the basic columns of $B^{-1}A$ form an identity, and

$$(A^\top p)_B = B^\top p = c_B, \quad r_B = c_B - B^\top p = 0.$$

Use: basic reduced costs vanish; complementary slackness at a basis; the exchange determinant.

M5. Unit vectors select

Me_j is column j of M and $e_i^\top M$ is row i . So $B^{-1}e_i$ is column i of B^{-1} , $e_\ell^\top B^{-1}b = x_{B(\ell)}$, and $e_\ell^\top B^{-1}A_j = u_\ell$. *Use:* reading B^{-1} from slack columns; extracting one tableau row as a vector $g^\top = e_\ell^\top B^{-1}$.

M6. Moving along a direction

If $Ad = 0$ then $A(x + \theta d) = Ax$ for every θ : the equations survive. For signs, with $\theta \geq 0$:

$$x + \theta d \geq 0 \iff \theta \leq \min_{j: d_j < 0} \frac{x_j}{-d_j} \quad (\text{no constraint if } d \geq 0).$$

If moreover $\text{supp}(d) \subseteq \text{supp}(x)$, then $x \pm \varepsilon d \geq 0$ for all small $\varepsilon > 0$: the move works in *both* directions.

M7. Dependence is a direction

The columns A_S are linearly dependent if and only if some $d \neq 0$ has $\text{supp}(d) \subseteq S$ and $Ad = 0$. Independent vectors of $\mathbb{R}^m$ number at most m , and when $\text{rank } A = m$ any independent set of columns extends, with further columns of A , to a basis.

M8. Compare coefficients of an identity

If $\alpha + \rho^\top x_N = \alpha' + \rho'^\top x_N$ for *every* x_N , then $\alpha = \alpha'$ (take $x_N = 0$) and $\rho = \rho'$ (take $x_N = e_j$). The same holds for affine functions of $x \in \mathbb{R}^n$. *Use:* the uniqueness of the tableau; reading

$y^* = -r_s$ from the final objective row; changing c .

M9. Replace one column of the identity

If F is the identity with column ℓ replaced by u , then $\det F = u_\ell$ (expand along row ℓ). So F is invertible exactly when $u_\ell \neq 0$. Since $B^{-1}B' = F$ when B' swaps A_j in for $A_{B(\ell)}$, the exchange gives a basis exactly when $u_\ell \neq 0$.

M10. Reformulate by an objective-preserving bijection

$x \mapsto (x, b - Ax)$ (slacks), $t = L + u$, $t = U - u$, $t = u - v$ (free), $\min f = -\max(-f)$. A map between feasible sets that preserves the objective transfers feasibility, boundedness, optimal solutions and the optimal value.

3 Eleven Proof Patterns

Each pattern has a trigger (“use when”), a template, and the places it already appeared.

P1. Insert the middle term

Use when comparing a primal quantity with a dual one. *Template:* $c^\top x \leq (A^\top y)^\top x = y^\top Ax \leq y^\top b$, using one sign condition on each side (M2, M3). *Seen in:* weak duality (canonical and general), the “not both” half of Farkas.

P2. Nonnegative terms summing to zero

Use when an equality must force many small conditions. *Template:* write the quantity as $\sum_k t_k$ with every $t_k \geq 0$; if the sum is 0, each $t_k = 0$. *Seen in:* complementary slackness (the gap is $\sum y_i s_i + \sum x_j r_j$), uniqueness when $r_N < 0$.

P3. Two-sided wiggle (contradict extremality)

Use when showing a point is *not* a vertex, or that a vertex has independent support. *Template:* from dependence get d with $Ad = 0$ and $\text{supp}(d) \subseteq \text{supp}(x)$; then $x \pm \varepsilon d \in P$ and x is their midpoint (M6, M7). *Seen in:* BFS = vertex ($\Leftrightarrow$).

P4. One-sided slide with the ratio test

Use when you want to reduce the support of a point without leaving P and without losing value. *Template:* orient d so that $c^\top d \geq 0$ and d has a negative entry; go to $x + \theta^* d$ with $\theta^* = \min_{d_j < 0} x_j / (-d_j)$. *Seen in:* purification, the simplex step, the dual simplex step.

P5. Induction on the support (or a minimal counterexample)

Use when a statement is obvious for points with independent support. *Template:* induct on $|\text{supp}(x)|$, or take a point with the *fewest* positive entries and show that dependence would give a point with fewer. *Seen in:* purification, Problem Set 1, Problem 6(c).

P6. An identity on the whole affine space

Use when a formula comes from pivots or substitutions. *Template:* the formula holds for every x with $Ax = b$, feasible or not, so you may plug in any such x , and compare coefficients (M8). *Seen in:* the tableau theorem, the optimality test, $y^* = -r_s$, the change of a basic profit.

P7. Improving direction (contrapositive)

Use when proving a *necessary* condition for optimality. *Template*: if the condition fails, exhibit a feasible direction with positive gain and enough room ($\theta_{\max} > 0$); this contradicts optimality. Room usually needs **nondegeneracy**. *Seen in*: “optimal and nondegenerate $\Rightarrow r_N \leq 0$ ”.

P8. Theorem of the alternative

Use when proving that something *exists* (a dual solution, a direction). *Template*: assume it does not exist, write that as an unsolvable linear system, let Farkas produce a certificate, and read the certificate as a contradiction with the hypotheses. *Seen in*: strong duality (the certificate (u, v, t) is a better plan if $t > 0$ and an improving ray if $t = 0$).

P9. Matching bounds

Use when proving that a *given* point is optimal. *Template*: exhibit a feasible point of the other problem with the same objective value; weak duality finishes. Never try to compare with “all” feasible points directly. *Seen in*: the matching-pair theorem, $p^\top = c_B^\top B^{-1}$ is dual optimal, every complementary slackness check.

P10. A maximum over a finite set

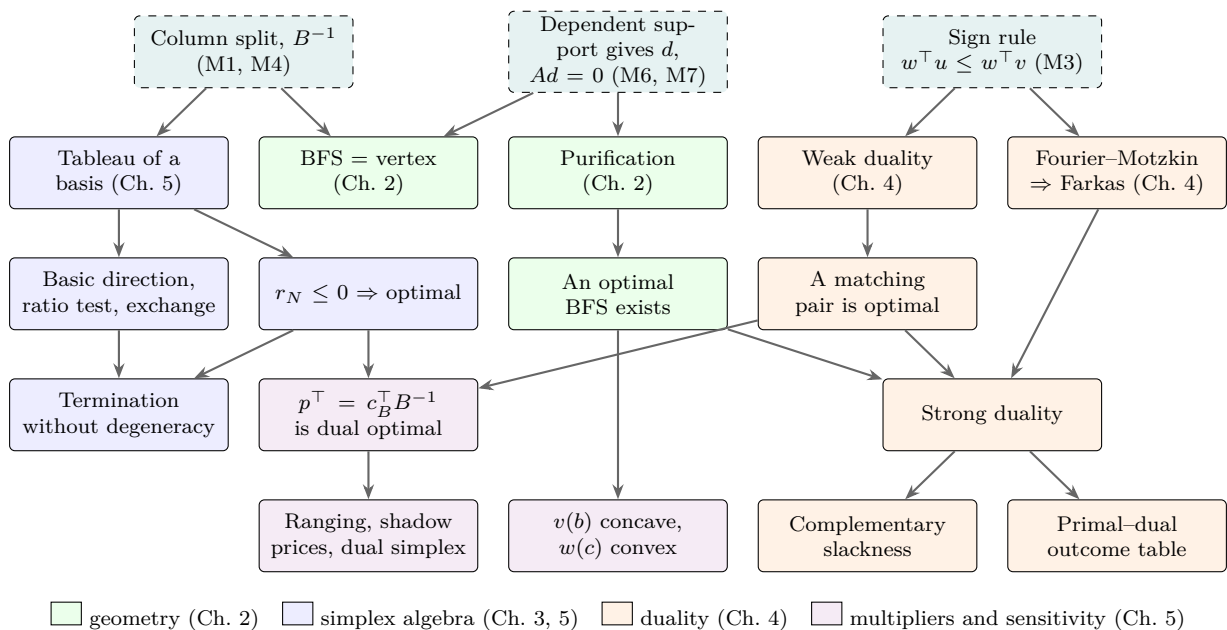
Use when an infinite search can be reduced to the finitely many bases. *Template*: show that every feasible point is beaten by some BFS; there are at most $\binom{n}{m}$ of them, and a maximum over a nonempty finite set exists. *Seen in*: the fundamental theorem, termination, $v(b) = \max_B c_B^\top B^{-1}b$.

P11. Combine solutions

Use when proving convexity, concavity, or that a set is a cone. *Template*: take solutions x_1, x_2 for two data sets, and check that $\lambda x_1 + (1 - \lambda)x_2$ (or $x_1 + x_2$, or tx_1) solves the combined data, using linearity. *Seen in*: the optimal set is convex; $v(b)$ is concave; $w(c)$ is convex.

4 The Theorem Map

The arrows say “is used to prove”. The three boxes on top are moves, not theorems; everything else is proved in the textbook chapters.



Every theorem in one line

For each result: the single idea of its proof, and the hypothesis that does the real work. If you can say the middle column out loud, you can write the proof.

Result	The idea of the proof	Hypothesis that matters
<i>Geometry (Chapters 1–2)</i>		
Feasible and optimal sets are convex	Combine two solutions (P11); a combination of optima has the same value	Linearity of $c^\top x$ and Ax
Canonical $\cong$ equational	Bijection $x \mapsto (x, b - Ax)$, objective unchanged (M10)	$s = b - Ax$ is unique
Basic solution well defined	$Ax = Bx_B + A_N x_N$; set $x_N = 0$; B invertible (M1)	B invertible
BFS $\Rightarrow$ vertex	Zeros of x_N force $u_N = w_N = 0$; then $u_B = w_B = B^{-1}b$	$0 < \lambda < 1$, $u, w \geq 0$
Vertex $\Rightarrow$ BFS	Dependence would give a wiggle $x \pm \varepsilon d$ (P3); extend the support to a basis (M7)	$\text{rank } A = m$ (to extend)
Purification	Orient d with $c^\top d \geq 0$ and a negative entry; ratio test kills one positive entry; induct (P4, P5)	Boundedness (forbids $d \geq 0$ with $c^\top d > 0$)
Fundamental theorem	Purify any point; take the best of finitely many BFSs (P10)	Feasible and bounded
<i>Simplex algebra (Chapters 3 and 5)</i>		
Tableau of a basis	Solve $Bx_B = b - A_N x_N$; substitute into $c^\top x$; uniqueness by $x_N = 0, e_j$ (M8)	B invertible
Basic direction	$A(x + \theta d) = b$ forces $Ad = 0$, so $Bd_B + A_j = 0$	$\theta > 0$
Ratio test, gain r_j	Coordinatewise signs (M6); $c^\top d = c_j - c_B^\top B^{-1}A_j = r_j$	$d_{B(k)} < 0$ rows only
Exchange is a basis	$B^{-1}B'$ is I with column ℓ replaced by u ; $\det = u_\ell \neq 0$ (M9)	$d_{B(\ell)} < 0$
$r_N \leq 0 \Rightarrow$ optimal	$c^\top u - c^\top x = \sum_{j \in N} r_j u_j \leq 0$ for every feasible u (P6, M3)	$u \geq 0$
Optimal $\Rightarrow r_N \leq 0$	Contrapositive: $r_j > 0$ gives an improving move (P7)	Nondegeneracy ($\theta_{\max} > 0$)
$r_N < 0 \Rightarrow$ unique	$\sum r_j u_j = 0$ forces $u_N = 0$ (P2), then $u_B = B^{-1}b$	Strict signs
Termination	Value strictly increases; finitely many bases (P10)	Nondegeneracy

Result	The idea of the proof	Hypothesis that matters
<i>Duality (Chapter 4)</i>		
Weak duality	Insert $y^\top Ax$ (P1)	$x \geq 0$ and $y \geq 0$, one each side
Matching pair optimal	Weak duality against every other feasible point (P9)	Both feasible
Fourier–Motzkin, Farkas	Pair every lower bound on x_1 with every upper bound; eliminate n times	Positive multipliers only
Strong duality	Farkas certificate (u, v, t) : $t > 0$ better plan, $t = 0$ improving ray (P8)	Primal feasible and bounded
Dual of the dual	Rewrite (D) as a canonical max and dualize	—
Complementary slackness	Gap = $\sum y_i s_i + \sum x_j r_j$, each term ≥ 0 (P2)	Strong duality for $(\Rightarrow)$
Outcome table	Weak duality rules out mixed cells; strong duality matches finite optima	Examples for the possible cells
<i>Multipliers and sensitivity (Chapter 5)</i>		
$p^\top = c_B^\top B^{-1}$ dual optimal	$r \leq 0$ is dual feasibility; $b^\top p = c_B^\top B^{-1}b$; then P9	B an optimal basis
Which test each change breaks	r and p do not involve b ; $B^{-1}b$ does not involve c	—
Shadow prices, b -ranging	$B^{-1}(b + \delta e_i) = x_B^* + \delta B^{-1}e_i \geq 0$; value $v + p^\top \Delta b$	Basis stays feasible
Basic profit change	Add $\varepsilon \times$ (row of $x_{B(k)}$) to the objective row (P6)	Uniqueness of the tableau
Dual simplex step	Ratio test on $(-r_j)/q_{\ell j}$ keeps $r \leq 0$; constant becomes positive	$\beta_\ell < 0$, $q_{\ell j^*} > 0$
v concave, w convex	Combine optimal solutions (P11); max of finitely many linear functions (P10)	Finite optimal values

5 Writing Proofs That Get Full Marks

Checklist.

1. **Write what is given and what must be shown**, with quantifiers: “Let $x \in P$ be arbitrary. We show $c^\top x \leq c^\top x^*$.”
2. **Use every hypothesis and say where**. If a hypothesis is never used, either the statement is stronger than you think or your proof has a gap. Rank m is used to extend independent columns; boundedness to orient a direction; nondegeneracy to get $\theta_{\max} > 0$; $x \geq 0$ in every sign rule.
3. **Do not assume attainment**. “Let x^* be optimal” is only allowed after the fundamental theorem (feasible and bounded) or strong duality has given you x^* .
4. **Prove both directions of an “if and only if”**, and label them $(\Rightarrow)$ and $(\Leftarrow)$.
5. **Prove optimality with a certificate**: exhibit the dual solution, check its feasibility, and check equal values.
6. **Counterexamples**: choose the smallest LP that works, and check every condition of the statement you are refuting.

Statements that look true but are false

- “A BFS has exactly m positive entries.” At most m ; fewer is degeneracy.
- “Different bases give different points.” Not at a degenerate vertex.

- “A positive reduced cost proves the BFS is not optimal.” Only if the BFS is nondegenerate.
- “A point with at most m positive entries is a BFS.” Its support columns must also be independent.
- “If the dual is infeasible, the primal is unbounded.” Both can be infeasible.
- “A shadow price is the value of any extra amount of the resource.” Only on the range where the basis stays optimal.
- “An unbounded feasible region means an unbounded objective.” The objective may still be bounded.

6 Problem Set: Algebra and Proofs

Rules. Give complete proofs. Any result of the textbook chapters may be quoted by name (for example “the tableau theorem”, “purification”, “strong duality”, “the outcome table”). A result proved in an earlier problem may be used in a later one. Unless stated otherwise, the setting is the equational form (E) with A of full row rank m , and (E^*) is its dual $\min\{b^\top p : A^\top p \geq c\}$. Stars give the difficulty. Full solutions are in *OR_Proof_Set_Solutions.pdf*.

Every problem is new: none repeats Problem Sets 1 to 4 or the textbook’s examples. Each one practises the moves (M) and patterns (P) of the review.

Part A. The algebra of bases and tableaux

A1. The Lagrangian identity and the dual of the equational form ★

- (a) Prove that for every $p \in \mathbb{R}^m$ and every $x \in \mathbb{R}^n$ with $Ax = b$,

$$c^\top x = b^\top p + (c - A^\top p)^\top x.$$

- (b) Use the dualization rules to show that the dual of (E) is (E^*) : $\min b^\top p$ s.t. $A^\top p \geq c$, p free. Then prove weak duality for this pair using only (a).
- (c) Let B be a basis and $p^\top = c_B^\top B^{-1}$. Show that $(A^\top p)_B = c_B$, so that $c - A^\top p$ is the reduced cost vector r and $r_B = 0$. Deduce the objective row of the tableau, $c^\top x = c_B^\top B^{-1}b + \sum_{j \in N} r_j x_j$, from (a).
- (d) Prove that B is an optimal basis ($B^{-1}b \geq 0$ and $r \leq 0$) if and only if its basic solution is feasible for (E) and p is feasible for (E^*) . Show that both are then optimal, with equal values.
- (e) Prove complementary slackness for this pair: feasible x and p are both optimal if and only if $x_j (A^\top p - c)_j = 0$ for every j . Explain why the basic solution and the multipliers of *any* basis satisfy these products, whether or not they are feasible.

A2. The pivot in matrix form ★★

Let B be a basis, $j \in N$, $u = B^{-1}A_j$, and suppose $u_\ell \neq 0$. Let B' be B with column ℓ replaced by A_j , and write $w = u - e_\ell$.

- (a) Show that $B^{-1}B' = F := I + w e_\ell^\top$.
- (b) Show that F is invertible with $F^{-1} = I - \frac{1}{u_\ell} w e_\ell^\top$, and deduce $B'^{-1} = F^{-1}B^{-1}$.
- (c) Describe what left multiplication by F^{-1} does to the rows of a matrix M . Explain why this is exactly a pivot of the simplex tableau on the entry u_ℓ .
- (d) Show that the multipliers and reduced costs of B' are

$$p'^\top = p^\top + \frac{r_j}{u_\ell} e_\ell^\top B^{-1}, \quad r'^\top = r^\top - \frac{r_j}{u_\ell} e_\ell^\top B^{-1}A.$$

Check that $r'_j = 0$ and $r'_{B(\ell)} = -r_j/u_\ell$, and interpret the sign of $r'_{B(\ell)}$ in a simplex pivot.

- (e) Check (d) on the running example of Chapter 5, $\max 3x + 2y$ s.t. $x + y + s_1 = 4$, $2x + y + s_2 = 6$, starting from the basis (s_1, x) with y entering.

A3. A degenerate vertex has several bases

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- (a) Prove: if x is a *nondegenerate* BFS, exactly one basis has x as its basic solution.
- (b) Show that the converse fails: give a degenerate BFS that belongs to exactly one basis.
- (c) Let x be the BFS of B with $x_{B(\ell)} = 0$, and suppose some $j \in N$ has $u_\ell = (B^{-1}A_j)_\ell \neq 0$. Prove that B' , obtained by replacing $A_{B(\ell)}$ with A_j , is a basis with the *same* basic solution x . Is any sign condition on u_ℓ needed?
- (d) Consider $\max 2x_1 + x_2$ s.t. $x_1 \leq 1$, $x_2 \leq 1$, $x_1 + x_2 \leq 2$, $x \geq 0$, with slacks s_1, s_2, s_3 .
 - (i) Prove that $(x_1, x_2) = (1, 1)$ is optimal with a one-line certificate.
 - (ii) Show that it is a degenerate BFS, and find every basis that has it as its basic solution.
 - (iii) Write the objective rows of the tableaux of $\{x_1, x_2, s_3\}$ and $\{x_1, x_2, s_1\}$. Which one certifies optimality? What does this say about the optimality test at a degenerate vertex?

A4. Splitting a free variable

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In an LP with a free variable x_1 , substitute $x_1 = u - v$ with $u, v \geq 0$, and write the result in equational form. The columns of u and v are A_1 and $-A_1$, with costs c_1 and $-c_1$.

- (a) Prove that no basis contains both u and v .
- (b) Deduce that every BFS has $uv = 0$, so that a vertex represents x_1 as $u = \max(x_1, 0)$, $v = \max(-x_1, 0)$.
- (c) Prove that $r_u + r_v = 0$ for every basis.
- (d) Deduce that at an optimal basis $r_u = r_v = 0$, that is, $(A^\top p)_1 = c_1$. Which dualization rule have you just recovered?

Part B. Polyhedra and vertices**B1. Optimal sets are faces**

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Let $S \subseteq \mathbb{R}^n$ be convex, let $c \in \mathbb{R}^n$, and suppose $v^* = \max_{x \in S} c^\top x$ is attained. Let $S^* = \{x \in S : c^\top x = v^*\}$.

- (a) Prove that S^* is convex.
- (b) Prove: if $z = \lambda u + (1 - \lambda)w$ with $u, w \in S$, $0 < \lambda < 1$ and $z \in S^*$, then $u, w \in S^*$.
- (c) Deduce that if S^* has two distinct points it has infinitely many, and that every extreme point of S^* is an extreme point of S .
- (d) Apply this to an LP in equational form with a finite optimum to give a *second* proof that some optimal solution is a BFS.

B2. A vertex is the unique optimum of some objective

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Prove that $\bar{x} \in P$ is a vertex of P if and only if there is $c \in \mathbb{R}^n$ such that $\bar{x}$ is the *unique* optimal solution of $\max\{c^\top x : x \in P\}$.

B3. A bounded P is the convex hull of its vertices

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Assume $P = \{x : Ax = b, x \geq 0\}$ is nonempty and *bounded*.

- (a) Let $x \in P$ have linearly dependent support columns, and let $d \neq 0$ satisfy $Ad = 0$ and

$\text{supp}(d) \subseteq \text{supp}(x)$. Prove that d has at least one positive and at least one negative entry.

- (b) Let $\theta_+ = \max\{\theta \geq 0 : x + \theta d \geq 0\}$ and $\theta_- = \max\{\theta \geq 0 : x - \theta d \geq 0\}$. Show that both are finite and positive, that $y = x + \theta_+ d$ and $z = x - \theta_- d$ lie in P and have smaller supports than x , and that

$$x = \frac{\theta_-}{\theta_+ + \theta_-} y + \frac{\theta_+}{\theta_+ + \theta_-} z.$$

- (c) Prove by induction on $|\text{supp}(x)|$ that every $x \in P$ is a convex combination of BFSs.
 (d) Deduce, *without* the fundamental theorem, that $\max\{c^\top x : x \in P\}$ is attained at a BFS.
 (e) Show that (c) fails without boundedness.

B4. Carathéodory for cones

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Let $A \in \mathbb{R}^{m \times n}$ be *any* matrix (no rank assumption) and suppose $b = Ax$ for some $x \geq 0$.

- (a) Prove that there is $x' \geq 0$ with $Ax' = b$ whose support columns $\{A_j : x'_j > 0\}$ are linearly independent. In particular, x' has at most $\text{rank } A \leq m$ positive entries.
 (b) A chemist must make a mixture with exact composition $b \in \mathbb{R}^m$ (amounts of m chemicals) from n stock solutions, one litre of solution j supplying A_j . Show that if it can be done at all, it can be done with at most m stock solutions.
 (c) Compare with purification: which hypothesis of purification is not needed here, and why?

Part C. Simplex theory

C1. Cycling forces degeneracy

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The simplex method of Chapter 5 produces bases $B_0, B_1, \dots, B_k$ with $B_k = B_0$ and $k \geq 1$. Let x^t be the BFS of B_t , let j_t be the entering index at step t (so $r_{j_t} > 0$), and let $\theta_t \geq 0$ be the step length.

- (a) Prove that $c^\top x^{t+1} - c^\top x^t = \theta_t r_{j_t} \geq 0$, and deduce that all the values $c^\top x^t$ are equal.
 (b) Deduce that every $\theta_t = 0$ and that $x^0 = x^1 = \dots = x^k$.
 (c) Prove that this common point is degenerate, in two ways: with the ratio test, and with Problem A3(a).
 (d) Conclude that if every BFS is nondegenerate, the method cannot cycle.

C2. When is the optimum unique?

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- (a) Let x^* be an optimal BFS of the basis B , and assume x^* is *nondegenerate*. Prove that x^* is the unique optimal solution if and only if $r_j < 0$ for every $j \in N$.
 (b) Show that the nondegeneracy hypothesis cannot be dropped: in the LP of Problem A3(d) with objective $\max x_1 + x_2$, show that $(1, 1)$ is the unique optimum but some optimal basis has a zero reduced cost.

C3. Certificates hidden in a tableau

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- (a) **Unboundedness.** At a BFS with basis B , suppose $r_j > 0$ and $d_B = -B^{-1}A_j \geq 0$. Show that $d = d^{(j)}$ satisfies $d \geq 0$, $Ad = 0$ and $c^\top d > 0$. Use this to prove *directly*, without the outcome table, that (E^*) is infeasible.

- (b) **Infeasibility.** A tableau row $x_{B(\ell)} = \beta_\ell + \sum_{j \in N} q_{\ell j} x_j$ has $\beta_\ell < 0$ and $q_{\ell j} \leq 0$ for every $j \in N$. Let $g^\top = e_\ell^\top B^{-1}$, row ℓ of B^{-1} . Show that $A^\top g \geq 0$ and $b^\top g < 0$, and prove directly that $P = \emptyset$.
- (c) In (b), assume also that every $r_j \leq 0$, as in the dual simplex method. Show that (E^*) is unbounded below, by exhibiting a ray of dual feasible points.

Part D. Duality and theorems of the alternative

D1. Three faces of Farkas ★★

Take as known the inequality form: exactly one of (i) $\exists x: Ax \leq b$, and (ii) $\exists y \geq 0: A^\top y = 0, b^\top y < 0$, holds.

- (a) **Equational form.** Prove that exactly one of the following holds: (i) $\exists x \geq 0$ with $Ax = b$; (ii) $\exists y$ with $A^\top y \geq 0$ and $b^\top y < 0$.
- (b) **Dual-feasibility form.** Prove that exactly one of the following holds: (i) $\exists p$ with $A^\top p \geq c$; (ii) $\exists d \geq 0$ with $Ad = 0$ and $c^\top d > 0$.
- (c) For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $b = (-1, 1)$, decide which alternative of (a) holds and give the certificate. Interpret (a) geometrically.

D2. Farkas from strong duality ★★

Assume strong duality for every primal–dual pair, and that the dual of the dual is the primal. Prove the equational Farkas lemma D1(a) from these, by looking at the LP $\max\{0^\top x : Ax = b, x \geq 0\}$ and its dual. Why does this show that strong duality and Farkas are “the same theorem”?

D3. A self-dual LP ★★

Let $K \in \mathbb{R}^{n \times n}$ be skew-symmetric ($K^\top = -K$), let $c \in \mathbb{R}^n$, and consider

$$(S) \quad \max c^\top x \quad \text{s.t.} \quad Kx \leq -c, x \geq 0.$$

- (a) Prove that $x^\top Kx = 0$ for every x .
- (b) Write the dual of (S) and show that it is $\min\{-c^\top y : Ky \leq -c, y \geq 0\}$, that is, minus (S) itself.
- (c) Prove that if (S) is feasible, its optimal value is 0, and every feasible x with $c^\top x = 0$ is optimal.
- (d) Check (c) for $K = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $c = (-1, 1)$.

D4. When is P bounded? ★★★

Let $P = \{x : Ax = b, x \geq 0\}$ be nonempty. Prove that P is bounded if and only if there is no $d \in \mathbb{R}^n$ with $d \geq 0, d \neq 0$ and $Ad = 0$. *Hint for ($\Leftarrow$): maximise each coordinate x_j over P and use D1(b).*

Part E. Value functions and sensitivity

E1. The value function $v(b)$

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For fixed A and c , let $P(b) = \{x : Ax = b, x \geq 0\}$, $F = \{b : P(b) \neq \emptyset\}$, and $v(b) = \max\{c^\top x : x \in P(b)\}$.

- Prove: if $v(b_0)$ is finite for one b_0 , then $v(b)$ is finite for every $b \in F$. *Hint: does the feasibility of (E^*) depend on b ?* Assume this from now on.
- Prove that F is a convex cone: $b_1, b_2 \in F$ and $t \geq 0$ imply $b_1 + b_2 \in F$ and $tb_1 \in F$.
- Prove that $v(tb) = tv(b)$ for $t > 0$, and $v(b_1 + b_2) \geq v(b_1) + v(b_2)$.
- Deduce that v is concave on F .
- Show that $v(0) = 0$. Interpret.

E2. Shadow prices are supergradients

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Keep the notation of E1, fix $b \in F$, and let p^* be an optimal solution of (E^*) at b .

- Prove that $v(b') \leq v(b) + p^{*\top}(b' - b)$ for every $b' \in F$.
- Deduce that for $\delta > 0$ (with $b \pm \delta e_i \in F$)

$$\frac{v(b + \delta e_i) - v(b)}{\delta} \leq p_i^* \leq \frac{v(b) - v(b - \delta e_i)}{\delta}.$$

- In the running example of Chapter 5 ($\max 3x + 2y, x + y \leq 4 + \delta, 2x + y \leq 6$), the value is $v = 10 + \delta$ on $[-1, 2]$ and $v = 12$ for $\delta \geq 2$. At $\delta = 2$, use (b) to show that every optimal dual solution has $0 \leq p_1 \leq 1$, and exhibit two optimal dual solutions with $p_1 = 1$ and $p_1 = 0$.
- State and prove the analogue of (a) for $w(c) = \max\{c^\top x : x \in P\}$, with an optimal x^* in the role of p^* .

E3. Optimality regions are convex

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- Fix an optimal basis B at (b, c) . Prove that $R = \{\Delta b : B \text{ is optimal for } b + \Delta b\}$ is a convex polyhedron containing 0, and that v is affine on $b + R$.
- Prove that $C_B = \{c : B \text{ is an optimal basis for } c\}$, with A and b fixed and $B^{-1}b \geq 0$, is a convex cone described by linear inequalities in c .
- Fix a feasible point x^* , not necessarily basic. Prove from the definition of optimality alone that $C(x^*) = \{c : x^* \text{ is optimal for } \max_{x \in P} c^\top x\}$ is a convex cone.
- For $\max c_1x + c_2y$ s.t. $x + y \leq 4, 2x + y \leq 6, x, y \geq 0$, prove that $(2, 2)$ is optimal if and only if $c = p_1(1, 1) + p_2(2, 1)$ for some $p \geq 0$.

Part F. True or false? Justify each answer

F. Eighteen statements

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- Every nonempty polyhedron $\{x \in \mathbb{R}^n : Ax \leq b\}$ has at least one vertex.
- Every nonempty $P = \{x : Ax = b, x \geq 0\}$ with $\text{rank } A = m$ has at least one vertex.
- Every BFS has exactly m positive entries.

4. If $x \in P$ has at most m positive entries, then x is a BFS.
5. Two different bases always have different basic solutions.
6. In a maximisation, if a BFS has a reduced cost $r_j > 0$, it is not optimal.
7. If an LP has a unique optimal solution, every optimal basis has $r_N < 0$.
8. The reduced costs of a basis do not depend on the sequence of pivots that reached it.
9. If x^* is a nondegenerate optimal BFS of (E), then (E*) has a unique optimal solution.
10. If (P) and (D) are both feasible, then both have optimal solutions.
11. If (P) is unbounded, then (D) is infeasible.
12. If (D) is infeasible, then (P) is unbounded.
13. If $Ax = b$, $x \geq 0$ has no solution, there is y with $A^\top y \geq 0$ and $b^\top y < 0$.
14. The optimal value $v(b)$ of a maximisation LP is a convex function of b .
15. A shadow price p_i gives the change in the optimal value for any change of b_i .
16. Changing the profit c_j of a nonbasic variable does not change the optimal value, as long as B stays optimal.
17. Adding a new variable to a maximisation LP can never decrease its optimal value.
18. If $v(b)$ is finite for one right-hand side b , the LP is not unbounded for any other right-hand side.